

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 6: DTFT and Fourier Series

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1 DTFT Pair (Recap)

DTFT and Inverse DTFT.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}, \quad (\text{Mitra 3.10})$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega. \quad (\text{Mitra 3.14})$$

$\Rightarrow x[n]$ is represented as a **mixture of infinitely many sinusoids** $e^{j\omega n}$.

1.1 DTFT and Fourier Series

For a periodic signal $s(t)$ with period T ,

$$s(t) = \sum_{n=-\infty}^{\infty} c_n e^{+j\frac{2\pi}{T}tn}.$$

Compare with

$$X(e^{j\omega}) = \underbrace{\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}}$$

Fourier series representation of the 2π -periodic function $X(e^{j\omega})$

i.e., the samples $x[n]$ play the role of the Fourier series coefficients of $X(e^{j\omega})$ (with period $T = 2\pi$ and the sign of the exponent flipped).

Duality.

- **DTFT:** discrete in time, periodic in frequency.
- **Fourier series:** periodic in time, discrete in frequency.

2 DTFT Properties

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2.1 Shift Properties

Property.

$$\begin{aligned} \text{Time shift:} \quad & y[n - M] \leftrightarrow e^{-j\omega M} Y(e^{j\omega}), \\ \text{Frequency shift:} \quad & y[n] e^{-j\theta_0 n} \leftrightarrow Y(e^{j(\omega + \theta_0)}). \end{aligned}$$

2.2 Convolution Theorems

Property.

$$\begin{aligned} \sum_{k=-\infty}^{\infty} y[k] x[n - k] &\leftrightarrow X(e^{j\omega}) Y(e^{j\omega}), \\ y[n] x[n] &\leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} Y(e^{j(\omega - \theta)}) X(e^{j\theta}) d\theta. \end{aligned}$$

¹cf. Mitra Sec. 3.2 and Table 3.4 (reproduced in the appendix of Lecture 5).

Proof (convolution).

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} y[k] x[n-k] \right) e^{-j\omega n} &= \sum_{k=-\infty}^{\infty} y[k] \left(\sum_{n=-\infty}^{\infty} x[n-k] e^{-j\omega n} \right) \\ &= \underbrace{\left(\sum_{k=-\infty}^{\infty} y[k] e^{-j\omega k} \right)}_{Y(e^{j\omega})} X(e^{j\omega}). \quad \blacksquare \end{aligned}$$

The inner sum in the first line is the DTFT of $x[n-k]$, which by the time-shift property is $e^{-j\omega k} X(e^{j\omega})$.

2.3 Parseval's Theorem

Property.

$$\sum_{n=-\infty}^{\infty} x[n] y^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) Y^*(e^{j\omega}) d\omega.$$

Note: setting $y[n] = x[n]$ gives the energy relation

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega.$$

Proof. Substitute the inverse DTFT $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} x[n] y^*[n] &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \right) y^*[n] \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) \left(\sum_{n=-\infty}^{\infty} y^*[n] e^{j\omega n} \right) d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) \underbrace{\left(\sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} \right)^*}_{Y^*(e^{j\omega})} d\omega. \quad \blacksquare \end{aligned}$$

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Example.

Find the DTFT of $x[n] = (n + 1) \alpha^n u[n]$, $|\alpha| < 1$.

We know

$$\underbrace{\alpha^n u[n]}_{y[n]} \leftrightarrow \underbrace{\frac{1}{1 - \alpha e^{-j\omega}}}_{Y(e^{j\omega})}, \quad \text{and} \quad x[n] = n y[n] + y[n].$$

Property (differentiation in frequency): $n y[n] \leftrightarrow j \frac{dY(e^{j\omega})}{d\omega}$. Hence

$$x[n] \leftrightarrow j \frac{dY(e^{j\omega})}{d\omega} + Y(e^{j\omega}) = \frac{1}{(1 - \alpha e^{-j\omega})^2}.$$

Remark

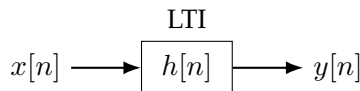
(addendum to orig lecture note).

The last step, written out:

$$j \frac{dY(e^{j\omega})}{d\omega} = j \cdot \frac{-j\alpha e^{-j\omega}}{(1 - \alpha e^{-j\omega})^2} = \frac{\alpha e^{-j\omega}}{(1 - \alpha e^{-j\omega})^2}, \quad Y(e^{j\omega}) = \frac{1 - \alpha e^{-j\omega}}{(1 - \alpha e^{-j\omega})^2},$$

and the two numerators add up to 1.

3 Frequency Response

**Note.**

From now on, our discussion will assume **zero ICs**, such that if $x[n] = 0$ for all n , then $y[n] = 0$ for all n . This removes the potential problem of having multiple responses corresponding to different ICs.

5 For an LTI system,

$$y[n] = x[n] \otimes h[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k].$$

If $x[n] = e^{j\omega n}$, $-\infty < n < \infty$:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} = \underbrace{e^{j\omega n}}_{x[n]} \underbrace{\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}}_{H(e^{j\omega})}.$$

$\Rightarrow e^{j\omega n}$ is an **eigenfunction** of the system (with eigenvalue $H(e^{j\omega})$).

Frequency Response.

$H(e^{j\omega})$ is the **frequency response** (the DTFT of the impulse response). In polar form,

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\theta(\omega)}.$$

Gain:²

$$\mathcal{G}(\omega) = 20 \log_{10} \left(\frac{|H(e^{j\omega})|}{|H(e^{j0})|} \right) \text{ dB}.$$

Attenuation: $-\text{Gain}$ (in dB).

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Example (Moving-Average Filter).

$$h[n] = \begin{cases} \frac{1}{M}, & 0 \leq n \leq M-1 \\ 0, & \text{else} \end{cases}$$

$$H(e^{j\omega}) = \frac{1}{M} \sum_{n=0}^{M-1} e^{-j\omega n} = \frac{1}{M} \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j(\frac{M-1}{2})\omega}$$

²As written in lecture, the gain is normalized by the DC value $|H(e^{j0})|$. Mitra (Sec. 4.8) defines the gain function without normalization, $\mathcal{G}(\omega) = 20 \log_{10} |H(e^{j\omega})|$ dB; the two differ only by a constant offset.

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(cf. the rectangular window in Lecture 5).

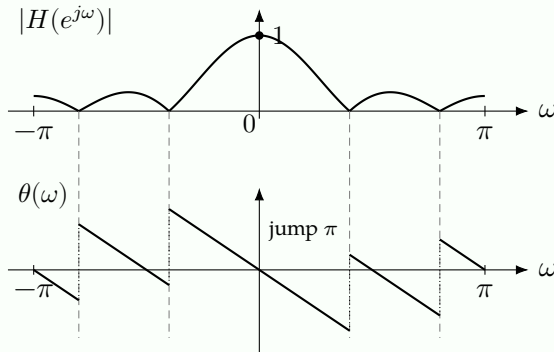
Magnitude response.

$$|H(e^{j\omega})| = \left| \frac{1}{M} \frac{\sin(\omega M/2)}{\sin(\omega/2)} \right|.$$

Phase response.

$$\theta(\omega) = -\frac{(M-1)}{2}\omega + \pi \sum_{k=1}^{\lfloor M/2 \rfloor} u\left(\omega - \frac{2\pi k}{M}\right), \quad 0 \leq \omega \leq \pi.$$

The jumps of π occur at the zeros $\omega = 2\pi k/M$ of the magnitude, where the real factor $\sin(\omega M/2)/\sin(\omega/2)$ changes sign.



Magnitude and phase of the moving-average filter for $M = 5$ (zeros at $\omega = \pm 2\pi/5, \pm 4\pi/5$).

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4 Linear Difference Equations and Frequency Response ³

$$\sum_{k=0}^N d_k y[n-k] = \sum_{q=0}^M p_q x[n-q].$$

³The handwritten page opens with “Phase is important!”—a teaser for the linear-phase discussion in Lecture 7.

Take the DTFT of both sides:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \left[\sum_{k=0}^N d_k y[n-k] \right] e^{-j\omega n} &= \sum_{n=-\infty}^{\infty} \left[\sum_{q=0}^M p_q x[n-q] \right] e^{-j\omega n} \\ \Rightarrow \sum_{k=0}^N d_k Y(e^{j\omega}) e^{-j\omega k} &= \sum_{q=0}^M p_q X(e^{j\omega}) e^{-j\omega q}. \end{aligned}$$

Frequency Response of an LDE.

For an LTI system, $Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega})$, so

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{q=0}^M p_q e^{-j\omega q}}{\sum_{k=0}^N d_k e^{-j\omega k}}.$$

Note.

Using this formula requires caution, because it is only applicable when the DTFT **converges** (cf. Lecture 5, Sec. *Convergence*).

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Example.

$$y[n] = \alpha y[n-1] + x[n], \quad |\alpha| < 1.$$

$$p_q = \{1\}_{\uparrow}, \quad d_k = \{1, -\alpha\}_{\uparrow}.$$

$$H(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}}, \quad |\alpha| < 1 \text{ (convergence).}$$

This agrees with the DTFT of $h[n] = \alpha^n u[n]$ computed in Lecture 5.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 3.6, 3.7, 4.6.4, 4.8, 4.9
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 2.7, 5.3